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Standard procedure of transformation

Genz (1992) presented a method for computing integrals, in which the change-of-variables step is particularly insightful. Applying this matrix decomposition and transformation approach to our design problem can simplify the integral to some extent. We first introduce Genz's standard transformation procedure.

A problem that arises in many statistics applications is that of computing the multivariate normal distribution function

$$F(\mathbf{a}, \mathbf{b}) = \frac{1}{\sqrt{|\Sigma|(2\pi)^m}} \int_{a_1}^{b_1} \int_{a_2}^{b_2} \dots \int_{a_m}^{b_m} e^{-\frac{1}{2}\theta^T \Sigma^{-1} \theta} d\theta,$$

where $\theta = (\theta_1, \theta_2, \dots, \theta_m)^T$ and Σ is an $m \times m$ symmetric positive definite covariance matrix.

According to the Cholesky decomposition, a positive definite covariance matrix can be decomposed as $\Sigma = \mathbf{C}\mathbf{C}^T$, where $\mathbf{C}$ is a lower triangular matrix. Applying the Cholesky transformation $\theta = \mathbf{C}\mathbf{y}$, we have $\theta^T \Sigma^{-1} \theta = \mathbf{y}^T \mathbf{C}^T (\mathbf{C}^T)^{-1} \mathbf{C}^{-1} \mathbf{C}\mathbf{y} = \mathbf{y}^T \mathbf{y}$, and $d\theta = |\mathbf{C}| d\mathbf{y} = |\Sigma|^{\frac{1}{2}} d\mathbf{y}$. Because $\mathbf{a} \leq \theta = \mathbf{C}\mathbf{y} \leq \mathbf{b}$ implies $(a_i - \sum_{j=1}^{i-1} c_{ij}y_j) / c_{ii} \leq y_i \leq (b_i - \sum_{j=1}^{i-1} c_{ij}y_j) / c_{ii}$ for $i = 1, 2, \dots, m$, we have

$$F(\mathbf{a}, \mathbf{b}) = \frac{1}{\sqrt{(2\pi)^m}} \int_{a'_1}^{b'_1} e^{-\frac{y_1^2}{2}} \int_{a'_2(y_1)}^{b'_2(y_1)} e^{-\frac{y_2^2}{2}} \dots \int_{a'_m(y_1, \dots, y_{m-1})}^{b'_m(y_1, \dots, y_{m-1})} e^{-\frac{y_m^2}{2}} d\mathbf{y},$$

with $a'_i(y_1, \dots, y_{i-1}) = (a_i - \sum_{j=1}^{i-1} c_{ij}y_j) / c_{ii}$ and $b'_i(y_1, \dots, y_{i-1}) = (b_i - \sum_{j=1}^{i-1} c_{ij}y_j) / c_{ii}$.

Application of the transformation to SSD design

To obtain a zero-mean vector, we define $\mathbf{u} = -\frac{1}{\sqrt{n}} \mathbf{Z}_{\mathcal{A}} \mathbf{C}_{\mathcal{A}}^{-1} \mathbf{F}_{\mathcal{A}}^T \mathbf{e}$. Then, $\mathbf{u}$ follows a k -dimensional normal distribution, denoted as $\mathbf{u} \sim N_k(\mathbf{0}, \mathbf{Z}_{\mathcal{A}} \mathbf{C}_{\mathcal{A}}^{-1} \mathbf{Z}_{\mathcal{A}})$, and the probability $P(S_{\lambda})$ can be expressed as

$$P(S_{\lambda}) = P\left(\mathbf{u} < \sqrt{n} \mathbf{Z}_{\mathcal{A}} \left(\mathbf{V}_{\mathcal{A}}^{\frac{1}{2}} \beta_{\mathcal{A}} - \lambda \mathbf{C}_{\mathcal{A}}^{-1} \mathbf{z}_{\mathcal{A}} \right)\right).$$

Denote $\mathbf{Z}_{\mathcal{A}} \mathbf{C}_{\mathcal{A}}^{-1} \mathbf{Z}_{\mathcal{A}}$ by Σ , and denote $\sqrt{n} \mathbf{Z}_{\mathcal{A}} (\mathbf{V}_{\mathcal{A}}^{\frac{1}{2}} \beta_{\mathcal{A}} - \lambda \mathbf{C}_{\mathcal{A}}^{-1} \mathbf{z}_{\mathcal{A}})$ by $\mathbf{v} = (v_1, v_2, \dots, v_k)^T$. Then

$$P(S_{\lambda}) = P(\mathbf{u} < \mathbf{v}) = \frac{1}{\sqrt{|\Sigma|(2\pi)^k}} \int_{-\infty}^{v_1} \int_{-\infty}^{v_2} \dots \int_{-\infty}^{v_k} e^{-\frac{1}{2}\mathbf{u}^T \Sigma^{-1} \mathbf{u}} d\mathbf{u}.$$

Applying the Cholesky decomposition to Σ , we have $\Sigma = \Gamma^{-1}(\Gamma^{-1})^T$, where both Γ^{-1} and Γ are lower triangular matrices and $\Gamma^{-1} = (L_{ij})_{k \times k}$. Let $\xi = \Gamma \mathbf{u}$, then ξ follows a k -dimensional standard normal distribution, i.e., $\xi \sim N_k(\mathbf{0}, \mathbf{I}_k)$. Now $\mathbf{u}^T \Sigma^{-1} \mathbf{u} = \xi^T (\Gamma^T)^{-1} \Gamma^T \Gamma \Gamma^{-1} \xi = \xi^T \xi$, and $d\xi = |\Gamma| d\mathbf{u} = |\Sigma|^{-\frac{1}{2}} d\mathbf{u}$. Because $\mathbf{u} = \Gamma^{-1} \xi \leq \mathbf{v}$ implies $\xi_i \leq (v_i - \sum_{j=1}^{i-1} L_{ij} \xi_j) / L_{ii}$ for $i = 1, 2, \dots, k$, we have

$$P(S_{\lambda}) = \frac{1}{\sqrt{(2\pi)^k}} \int_{-\infty}^{v'_1} \int_{-\infty}^{v'_2} \dots \int_{-\infty}^{v'_k} e^{-\frac{1}{2}\xi^T \xi} d\xi = \frac{1}{\sqrt{(2\pi)^k}} \int_{-\infty}^{v'_1} e^{-\frac{\xi_1^2}{2}} \int_{-\infty}^{v'_2(\xi_1)} e^{-\frac{\xi_2^2}{2}} \dots \int_{-\infty}^{v'_k(\xi_1, \dots, \xi_{k-1})} e^{-\frac{\xi_k^2}{2}} d\xi,$$

with $v'_i(\xi_1, \dots, \xi_{i-1}) = (v_i - \sum_{j=1}^{i-1} L_{ij} \xi_j) / L_{ii}$.

Subsequent treatment

(1)

Since the density has been reduced to the standard normal distribution, we focus on the integration region. In the nested integration of Genz (1992), we have $a'_i(y_1, \dots, y_{i-1}) = (a_i - \sum_{j=1}^{i-1} c_{ij}y_j)/c_{ii}$ and $b'_i(y_1, \dots, y_{i-1}) = (b_i - \sum_{j=1}^{i-1} c_{ij}y_j)/c_{ii}$. We note that the length of the i -th integration interval is $b'_i(y_1, \dots, y_{i-1}) - a'_i(y_1, \dots, y_{i-1}) = (b_i - a_i)/c_{ii}$, which is independent of the first $i-1$ random variables, thus convenient to handle. In our problem, $b_i = v_i$, $c_{ij} = L_{ij}$, and all a_i are identically $-\infty$; therefore, we initially thought that we might only consider the quantities $\frac{b_i}{L_{ii}}$. To comprehensively account for the regions corresponding to all components, we considered adopting the criterion

$$\operatorname{argmax}_{\mathbf{C}_{\mathcal{A}}} \prod_{i=1}^k \frac{v_i}{L_{ii}} = \operatorname{argmax}_{\mathbf{C}_{\mathcal{A}}} \frac{\prod_{i=1}^k v_i}{|\Gamma|}$$

or the heuristic criterion

$$\operatorname{argmax}_{\mathbf{C}_{\mathcal{A}}} \frac{\mathbf{v}^T \mathbf{1}}{|\Gamma|} = \operatorname{argmax}_{\mathbf{C}_{\mathcal{A}}} |\Gamma|^{-1} \mathbf{v}^T \mathbf{1} = \operatorname{argmax}_{\mathbf{C}_{\mathcal{A}}} |(\mathbf{Z}_{\mathcal{A}} \mathbf{C}_{\mathcal{A}}^{-1} \mathbf{Z}_{\mathcal{A}})^{-1}| \left(\mathbf{Z}_{\mathcal{A}} \left(\mathbf{V}_{\mathcal{A}}^{\frac{1}{2}} \beta_{\mathcal{A}} - \lambda \mathbf{C}_{\mathcal{A}}^{-1} \mathbf{z}_{\mathcal{A}} \right) \right)^T \mathbf{1}.$$

We had originally intended to perform matrix differentiation, explore whether explicit solutions exist in the one- or two-parameter cases, examine the first and second derivatives, and investigate the extremum points and local maxima.

(2)

However, we have found that the above derivation and criterion are not rigorous.

For the normal distribution, the interval length has no practical meaning. Two intervals of the same length located at different positions can yield vastly different integral values; in particular, when the midpoint lies at the origin, the integral value is maximized. The integral kernel of the Genz (1992) algorithm is $\Phi(b'_i) - \Phi(a'_i)$, not $b'_i - a'_i$. Since our lower bounds are all $-\infty$, the probability is equivalent to $\Phi(b'_i)$, but the numerator of b'_i still contains the term $\sum_{j=1}^{i-1} c_{ij}y_j$. If we forcibly take differences and ignore the summation term, we are effectively eliminating the correlation artificially. If we intend to differentiate to study the design properties, the chain rule must be applied to $\Phi(b'_i)$ and the y_j 's inside the summation are integration variables, which cannot be treated as constants and canceled out.

In our SSD design problem, it is preferable to focus on the integration upper limits $\mathbf{v}' = (v'_1, v'_2, \dots, v'_k)^T$. We know that

$$\mathbf{v} = (v_1, v_2, \dots, v_k)^T = \sqrt{n} \mathbf{Z}_{\mathcal{A}} \left(\mathbf{V}_{\mathcal{A}}^{\frac{1}{2}} \beta_{\mathcal{A}} - \lambda \mathbf{C}_{\mathcal{A}}^{-1} \mathbf{z}_{\mathcal{A}} \right) \quad \text{and} \quad v'_i(\xi_1, \dots, \xi_{i-1}) = \left(v_i - \sum_{j=1}^{i-1} L_{ij} \xi_j \right) / L_{ii}.$$

(i) Since the term $-\sum_{j=1}^{i-1} L_{ij} \xi_j$ in v'_i is rather complicated, one possible approach is to take $\mathbb{E}_{\xi}[v'_i] = \frac{v_i}{L_{ii}}$. However, we do not recommend using a product-type criterion here. First, there is no guarantee that all expected upper limits are positive, which may lead to substantial cancellation of positive and negative terms. Second, the upper limits themselves do not directly determine the integral value; for instance, in the two-dimensional case, the integral value when the upper limits are $(10\sqrt{10}, 10\sqrt{10})$ is far greater than when they are $(1, 1000)$. We consider the criterion

$$\operatorname{argmax}_{\mathbf{C}_{\mathcal{A}}} \left(\min_i \mathbb{E}_{\xi}[v'_i] \right) = \operatorname{argmax}_{\mathbf{C}_{\mathcal{A}}} \left(\min_i \frac{v_i}{L_{ii}} \right),$$

which might be effective, but the maximin criterion still only takes into account the smallest upper-limit component. In the future, we may employ some methods to reduce the search effort and evaluate the effectiveness of this criterion via numerical simulations.

(ii) Another attempt is to take expectations of the CDF. Since

$$\mathbb{E}_{\xi}[v'_i] = \frac{v_i}{L_{ii}}, \quad \operatorname{Var}_{\xi}[v'_i] = \frac{\sum_{j=1}^{i-1} L_{ij}^2}{L_{ii}^2},$$

we have

$$\mathbb{E}_\xi[\Phi(v'_i)] = \Phi\left(\frac{\mathbb{E}_\xi[v'_i]}{\sqrt{1 + \text{Var}_\xi[v'_i]}}\right) = \Phi\left(\frac{v_i}{\sqrt{\sum_{j=1}^i L_{ij}^2}}\right).$$

We might adopt the criterion

$$\prod_{i=1}^k \mathbb{E}_\xi[\Phi(v'_i)] = \prod_{i=1}^k \Phi\left(\frac{v_i}{\sqrt{\sum_{j=1}^i L_{ij}^2}}\right),$$

but this does not reflect the true nature of the nested integral. In the nested integration, the integrand of the second-from-innermost layer is already $\phi(\xi_{k-1})\Phi(v'_k)$, and $\Phi(v'_{k-1})$ does not appear directly in the integral. As long as there is correlation between layers, the result will deviate from the correct one. We have constructed a two-dimensional example where this criterion deviates severely. Therefore, even if we can continue to study this criterion analytically and numerically in the future, we are not very optimistic about it. The criterion

$$\operatorname{argmax}_{\mathbf{C}_{\mathcal{A}}} \left(\min_i \prod_{i=1}^k \mathbb{E}_\xi[\Phi(v'_i)] \right) = \operatorname{argmax}_{\mathbf{C}_{\mathcal{A}}} \left(\min_i \prod_{i=1}^k \Phi\left(\frac{v_i}{\sqrt{\sum_{j=1}^i L_{ij}^2}}\right) \right)$$

is worth trying. This is also a maximin criterion, and it may be more effective than simply taking expectations of $\mathbb{E}_\xi[v'_i]$. At present, it seems difficult to find a reasonable, accurate, and analytically tractable criterion; we are still stuck at the level of search and simulation.